

In some situations, however, the covariate x is observed with error and the observed value is

$$X_i = x_i + u_i, \quad (1.2)$$

where $u_i \stackrel{iid}{\sim} N(0, \sigma_u^2)$, $i = 1, \dots, n$ (*iid*: independent and identically distributed). In (1.1) and (1.2), e_i and u_i are considered to be independent. When the covariate x is a fixed quantity, the functional model follows. If x is a random variable, the structural model results (Fuller, 1987, Section 1.1). This paper is devoted to the functional model, with one of the variances, σ_u^2 , known. By considering the functional version, unnecessary assumptions about the distribution of the covariate x are avoided.

The linear calibration problem encompasses two steps. In the first step (calibration stage), the linear regression model parameters are estimated. In the second step, a random sample of size k of the dependent variable y corresponding to an unknown value of x , which we denote by x_0 , is observed. For this step, we have

$$y_{0j} = \alpha + \beta x_0 + e_{0j}, \quad (1.3)$$

where, as in the first stage, $e_{0j} \stackrel{iid}{\sim} N(0, \sigma_e^2)$, $j = 1, \dots, k$. The calibration problem for the ordinary regression model has been extensively studied in the literature. Bayesian and classical estimation procedures abound in the literature. The main results are summarized in Brown (1993). Two main estimators are considered in the literature for x_0 in the ordinary regression model. The classical, or maximum likelihood estimator (MLE) and the inverse estimator, obtained by regressing x on y . A Bayesian justification for the inverse estimator is given in Hoadley (1970). Shukla (1972) derives first order approximation to the asymptotic biases and mean squared errors of the two estimators, under the assumption that $|\hat{\beta}| > 0$. Comparisons between the

asymptotic bias and mean squared error conducted by Shukla (1972) show that the inverse estimator is preferable when the observable mean $\bar{X} = \sum_{i=1}^n X_i/n$ is close to the unknown x_0 . The overall better performance of the classical estimator, however, is indicated by Shukla (1972) for its properties in large samples and also in the absence of any prior information about the unknown x_0 . Fuller (1987, Section 2.5) discusses properties of the inverse estimator in the functional model. Some comparisons between the classical and the inverse estimators in the regression model with measurement errors specified by (1.1)–(1.3) with $\lambda = \sigma_e^2/\sigma_u^2$ known and $k = 1$ are reported in Lee and Yum (1989). They derive approximate expression for the asymptotic mean squared error of the classical and inverse estimators in the functional measurement error model (1.1)–(1.3) with several combinations of the parameter estimates. As shown in Bolfarine et al. (1997), however, their asymptotic mean squared error expressions do not involve all the n^{-1} terms, as considered by Shukla (1972) in the ordinary regression model.

In this paper, n^{-1} approximations are obtained for the asymptotic mean squared error of the classical and inverse estimators, with several combinations of the parameter estimates. As is well known, the least squares estimators of α and β are not consistent under the measurement error model given by (1.1)–(1.2). To make the estimation problem feasible, additional assumptions are required. This paper discusses the situation where one of the variances, σ_u^2 , is considered as known. It can be shown that similar results hold for the case σ_e^2 known.

Section 2 discusses large sample properties of the naive estimators. Section 3 considers consistent estimators for the model specified by equations (1.1)–(1.3) and first order approximations are obtained for their means and variances.

Section 4 presents some analytical and numerical comparisons between the asymptotic mean squared errors of the estimators considered. Section 5 is dedicated to an application with a real data set. The main conclusion when comparing the estimators considered is that the classical estimator computed with the consistent estimators of the parameters α and β behaves better, in terms of asymptotic mean squared error than the other estimators. This conclusion seems to take effect specially when the average of the unobserved x_i (and consequently that of X_i) values is far from x_0 .

2. Large sample properties of the naive estimators

As mentioned in the introduction, to estimate x_0 we can consider the classical and inverse estimators in the model defined by (1.1) and (1.3) when x is measured without error ($\sigma_u^2 = 0$). These estimators can be written as

$$\hat{x}_L = \frac{\bar{y}_0 - \hat{\alpha}_L}{\hat{\beta}_L} \quad \text{and} \quad \tilde{x}_L = \tilde{\alpha}_L + \tilde{\beta}_L \bar{y}_0, \quad (2.1)$$

where $\hat{\alpha}_L = \bar{y} - \bar{x}\hat{\beta}_L$, $\hat{\beta}_L = S_{xy}/S_{xx}$, $\tilde{\alpha}_L = \bar{x} - \bar{y}\tilde{\beta}_L$ and $\tilde{\beta}_L = S_{xy}/S_{yy}$, $\bar{y} = \sum_{i=1}^n y_i/n$, $\bar{x} = \sum_{i=1}^n x_i/n$, $\bar{y}_0 = \sum_{j=1}^k y_{0j}/k$, $S_{yy} = \sum_{i=1}^n (y_i - \bar{y})^2/n$, $S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2/n$, $S_{xy} = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})/n$. As shown in Shukla (1972),

$$\hat{x}_L \xrightarrow{P} x_0 \quad \text{and} \quad \tilde{x}_L \xrightarrow{P} x_0 + \frac{\sigma_e^2 (\bar{x} - x_0)}{\beta^2 \nu^2 \theta}$$

where $\theta = 1 + \sigma_e^2/\beta^2 \nu^2$ and $\lim_{n \rightarrow \infty} S_{xx} = \nu^2$. Thus, $\hat{x}_L$ is consistent and $\tilde{x}_L$ is not, unless $\bar{x} = x_0$. We may now ask the question: What happens with the naive estimators $\hat{x}_L$ and $\tilde{x}_L$ defined by (2.1) when x is measured with error? In this case we observe X as specified by (1.2) and not the true x , which remains unknown. To answer this question we consider first the large sample behaviour of $\hat{\beta}_L$ and $\tilde{\beta}_L$, with x_i replaced by the

observable $X_i, i = 1, \dots, n$. We remark that the distributions of $\hat{\beta}_L$ and $\tilde{\beta}_L$ have no finite moments (Kendall and Stuart, 1952) so that the results presented are approximations to the moments of the asymptotic distributions of those estimators, which exist. Davies and Hulton (1975) report also some studies on the asymptotic bias and mean squared error of the least squares estimator $\hat{\beta}_L$. They emphasize, however, only the leading terms of the approximations. However, results for $\tilde{\beta}_L$ are not considered in the literature.

Lemma 2.1. *Under the model specified by (1.1)–(1.2) with σ_u^2 known, it follows that the mean and variance of $\hat{\beta}_L$ and $\tilde{\beta}_L$, up to first order terms, are given by*

$$\begin{aligned} E[\hat{\beta}_L] &= \frac{\beta l}{1+l} \left(1 + \frac{1+3l}{n(1+l)^2}\right), \\ \text{Var}[\hat{\beta}_L] &= \frac{\sigma_e^2}{n\sigma_u^2(1+l)} + \frac{\beta^2 l(1+l^2)}{n(1+l)^4}, \\ E[\tilde{\beta}_L] &= \frac{Z}{\beta(1+Z)} + \frac{Z(1+3Z)}{n\beta(1+Z)^3} \end{aligned}$$

and

$$\text{Var}[\tilde{\beta}_L] = \frac{\sigma_u^2}{n\sigma_e^2(1+Z)} + \frac{Z(1+Z^2)}{n\beta^2(1+Z)^4},$$

where

$$l = S_{xx}/\sigma_u^2 \quad \text{and} \quad Z = \beta^2 \sigma_u^2 l / \sigma_e^2 = \beta^2 S_{xx} / \sigma_e^2.$$

Proof. See Appendix.

Notice from Lemma 2.1 that $\hat{\beta}_L$ is not consistent for β , since, as $n \rightarrow \infty$, $\hat{\beta}_L \xrightarrow{P} \frac{\nu^2}{\nu^2 + \sigma_u^2} \beta$. Thus, $\hat{\beta}_L$ is attenuated toward zero. We present next the large sample behaviour of $\hat{x}_L$ and $\tilde{x}_L$, under the measurement error model (1.1)–(1.3), computed with x_i replaced by $X_i, i = 1, \dots, n$.

Theorem 2.1. *Under the model specified by (1.1)–(1.3), with k fixed and given that $|\hat{\beta}| > 0$, first order approximations for the mean and variance of*

$\hat{x}_L$ and $\tilde{x}_L$, are given by

$$E[\hat{x}_L] = x_0 + \frac{(x_0 - \bar{x})}{l} + \frac{(x_0 - \bar{x})}{nl} \left[\frac{(1+l^2)}{l(1+l)} + \frac{\sigma_e^2(1+l)^2}{\sigma_u^2 \beta^2 l^2} - \frac{(1+3l)}{1+l} \right],$$

$$Var[\hat{x}_L] = \frac{1}{n} \left(\sigma_u^2 + \frac{\sigma_e^2(1+l)^2}{\beta^2 l^2} \right) + \frac{\sigma_e^2}{k\beta^2 l^2} (1+l)^2 - \frac{2\sigma_e^2}{nk\beta^2 l^2} (1+3l)$$

$$+ ((x_0 - \bar{x})^2 + \frac{3\sigma_e^2}{k\beta^2}) \left[\frac{\sigma_e^2(1+l)^3}{n\beta^2 l^4 \sigma_u^2} + \frac{(1+l^2)}{nl^3} \right],$$

$$E[\tilde{x}_L] = x_0 - \frac{(x_0 - \bar{x})}{(1+Z)} + \frac{(x_0 - \bar{x})Z(1+3Z)}{n(1+Z)^3}$$

and

$$Var[\tilde{x}_L] = \frac{1}{n} \left(\sigma_u^2 + \frac{\sigma_e^2 Z^2}{\beta^2 (1+Z)^2} \right) + \frac{\sigma_e^2 Z^2}{k\beta^2 (1+Z)^2}$$

$$+ \frac{(x_0 - \bar{x})^2}{n(1+Z)} \left(\frac{\beta^2 \sigma_u^2}{\sigma_e^2} + \frac{Z(1+Z^2)}{(1+Z)^3} \right)$$

$$+ \frac{1}{nk(1+Z)} \left(\sigma_u^2 + \frac{\sigma_e^2 Z(7Z^2 + 2Z + 1)}{\beta^2 (1+Z)^3} \right),$$

with l and Z as given in Lemma 2.1.

Proof. See Appendix.

Thus, from the above theorem, it follows that

$$\hat{x}_L \xrightarrow{P} x_0 + \frac{(x_0 - \bar{x})}{l} \quad \text{and} \quad \tilde{x}_L \xrightarrow{P} x_0 - \frac{(x_0 - \bar{x})}{(1+Z)},$$

meaning that $\hat{x}_L$ and $\tilde{x}_L$ are not consistent, unless $\bar{x} = x_0$. Moreover, $\hat{x}_L$ is unbounded with respect to σ_u^2 , that is, $|\hat{x}_L| \rightarrow \infty$ as $\sigma_u^2 \rightarrow \infty$, which is the opposite to the attenuation effect that affects $\hat{\beta}_L$. On the other hand, the limit of $\tilde{x}_L$ does not depend on σ_u^2 and so its bias is not affected by σ_u^2 . Such peculiarities of $\hat{x}_L$ and $\tilde{x}_L$ are also noted in the simulation study that we present in Section 4. However, when n is large and $\bar{X}$ is close to x_0 , we expect that the large sample biases of $\hat{x}_L$ and $\tilde{x}_L$ are small.

3. Consistent estimators and large sample properties

As shown in Moberg and Sundberg (1978), when σ_u^2 is known, the maximum likelihood estimators of α and β are not consistent. This does not mean, however, that consistent estimators cannot be obtained. A common procedure used in functional regression models to obtain estimators for α and β , is to adopt the estimators obtained by using the method of moments in the structural model and then studying its properties in the functional case (see, for example, Cheng and VanNess, 1991). So, we use the same approach to obtain consistent estimators in the functional linear calibration model when σ_u^2 is known.

From (1.1)–(1.3) it follows that

$$\begin{pmatrix} X_i \\ y_i \end{pmatrix} \sim N\left(\begin{pmatrix} x_i \\ \alpha + \beta z_i \end{pmatrix}; \begin{pmatrix} \sigma_u^2 & 0 \\ 0 & \sigma_e^2 \end{pmatrix}\right) \quad (3.1)$$

$i = 1, \dots, n$, and

$$y_{0j} \sim N(\alpha + \beta x_0, \sigma_e^2), \quad (3.2)$$

$j = 1, \dots, k$. The structural model will be considered to obtain consistent estimators for the parameters of the model (3.1) and (3.2). Suppose that σ_u^2 is known and that $x_i \sim N(\mu_x, \sigma_x^2)$, $i = 1, \dots, n$. Under these assumptions it can be shown that the population moments satisfy:

$$(\sigma_X^2, \sigma_y^2, \sigma_{y_0}^2, \sigma_{Xy}, \sigma_{Xy_0}, \sigma_{yy_0}) = (\sigma_x^2 + \sigma_u^2, \beta^2 \sigma_x^2 + \sigma_e^2, \sigma_e^2, \beta \sigma_x^2, 0, 0)$$

and

$$(\mu_X, \mu_y, \mu_{y_0}) = (\mu_x, \alpha + \beta \mu_x, \alpha + \beta x_0).$$

The estimators of the unknown parameters are obtained by replacing the unknown population moments by the corresponding sample moments in the left-hand side of the above expressions. Thus, a system of equations expressed as functions of the unknown parameters is obtained. By solving this system,

the following estimators are obtained:

$$\hat{\alpha} = \bar{y} - \hat{\beta}\bar{X}, \quad \hat{\beta} = \frac{S_{Xy}}{S_{XX} - \sigma_u^2}, \quad (3.3)$$

$$\hat{\sigma}_e^2 = \frac{(S_{yy} + S_{yy}^0 - \hat{\beta}S_{Xy})}{2}, \quad (3.4)$$

$$\hat{x} = \frac{\bar{y}_0 - \hat{\alpha}}{\hat{\beta}} \quad (3.5)$$

and

$$\hat{\sigma}_x^2 = S_{XX} - \sigma_u^2, \quad (3.6)$$

where $\bar{y} = \sum_{i=1}^n y_i/n$, $\bar{X} = \sum_{i=1}^n X_i/n$, $\bar{y}_0 = \sum_{j=1}^k y_{0j}/k$, $S_{yy} = \sum_{i=1}^n (y_i - \bar{y})^2/n$, $S_{XX} = \sum_{i=1}^n (X_i - \bar{X})^2/n$, $S_{Xy} = \sum_{i=1}^n (X_i - \bar{X})(y_i - \bar{y})/n$ and $S_{yy}^0 = \sum_{j=1}^k (y_{0j} - \bar{y}_0)^2/k$.

The estimators given in (3.3)–(3.6) are well defined in the parametric space if $\hat{\sigma}_e^2$ and $\hat{\sigma}_x^2$ are positive. These estimators are positive if $S_{XX} > \sigma_u^2$ and $S_{yy} + S_{yy}^0 > S_{Xy}^2/(S_{XX} - \sigma_u^2)$. Asymptotically, there are no restrictions because $\hat{\sigma}_x^2 \rightarrow \sigma_x^2 > 0$ and $\hat{\sigma}_e^2 \rightarrow \sigma_e^2 > 0$.

As pointed out by a referee (see also (3.3)), $\hat{x}$ is expected to present better behaviour in situations where the measurement error variance σ_u^2 is smaller than the spread of the auxiliary variable X . In practical applications, such cases may arise in controlled calibration problems where the variable X is controlled but subject to small or moderate measurement errors due to complexity of modern instrumentation.

Considering the functional model, the strong consistency of the estimators given in (3.3)–(3.6) is proved under the assumption that

$$\lim_{n \rightarrow \infty} \bar{x} = \mu \quad \text{and} \quad \lim_{n \rightarrow \infty} S_{xx} = \nu^2, \quad (3.7)$$

are both finite, where $\bar{x} = \sum_{i=1}^n x_i/n$ and $S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2/n$.

Theorem 3.1. *Under the measurement error model defined by (1.1)–(1.3) and assumptions (3.7), with σ_u^2 known, $k = O(n)$ in (1.3), it follows that the estimators $\hat{\alpha}, \hat{\beta}, \hat{x}$ and $\hat{\sigma}_e^2$ are strongly consistent.*

Proof. See Appendix.

First order approximations for the bias and mean squared error of the classical and inverse estimators are considered next. A lemma is presented first, which is used for proving the main result. As in Lemma 2.1, the results are first order approximators to the asymptotic distributions.

Lemma 3.1. *Under the model specified by (1.1)–(1.2) it follows, up to terms of order $O(n^{-1})$, that*

$$\begin{aligned} E[\hat{\beta}] &= \beta + \frac{\beta\sigma_u^2}{nS_{xx}} \left(3 + \frac{2\sigma_u^2}{S_{xx}}\right), \\ \text{Var}[\hat{\beta}] &= \frac{p}{n}, \end{aligned} \quad (3.8)$$

where

$$p = \frac{[(\sigma_u^2 + S_{xx})\varphi + \beta^2\sigma_u^4]}{S_{xx}^2} \quad \text{and} \quad \varphi = \sigma_e^2 + \beta^2\sigma_u^2.$$

Proof. See Appendix.

In the next theorem, approximations up to terms of order $O(n^{-1})$ are presented for the mean and variance of $\hat{x}$ given that $|\hat{\beta}| > 0$, as considered in Shukla (1972).

The proof of the next theorem is similar to that of Theorem 2.1 and so it is not presented.

Theorem 3.2. *Under the model specified by (1.1)–(1.3), with σ_u^2 known and k fixed, given that $|\hat{\beta}| > 0$, it follows, up to terms of order $O(n^{-1})$, that*

$$E[\hat{x}] = x_0 + \frac{(x_0 - \bar{x})[(\sigma_e^2 - 2\beta^2\sigma_u^2) + \sigma_u^2\sigma_e^2/S_{xx}]}{n\beta^2S_{xx}}, \quad (3.9)$$

$Var[\hat{x}] = \frac{\varphi}{n\beta^2} + \frac{\sigma_e^2}{k\beta^2} - \frac{2\sigma_e^2\sigma_u^2}{nk\beta^2 S_{xx}} [3S_{xx} + 2\sigma_u^2] + \left((x_0 - \bar{x})^2 + \frac{3\sigma_e^2}{k\beta^2} \right) \frac{p}{n\beta^2}$,
with φ and p as given in Lemma 3.1.

We call attention to the fact that if $\sigma_u^2 = 0$, then $Var[\hat{x}]$ coincides with the corresponding expression given in Shukla (1972). The next theorem presents a consistent estimator for the asymptotic mean squared error of $\hat{x}$. Its proof follows directly from results presented earlier.

Theorem 3.3. *Under the assumptions considered in Theorem 3.2, a consistent estimator for the asymptotic mean squared error of $\hat{x}$ is given by*

$$\widehat{MSE}[\hat{x}] = \frac{\hat{\varphi}}{n\hat{\beta}^2} + \frac{\hat{\sigma}_e^2}{k\hat{\beta}^2} + \frac{\hat{\sigma}_e^2(2 - 3\hat{l})}{nk\hat{\beta}^2\hat{l}^2} \\ + (\hat{x} - \bar{X})^2 \left(\frac{\hat{\varphi}(\hat{l} + 1) + \hat{\beta}^2\sigma_u^2}{n\hat{\beta}^2\hat{l}^2\sigma_u^2} \right) + \frac{3\hat{\sigma}_e^4(1 + \hat{l})}{nk\hat{\beta}^4\hat{l}^2\sigma_u^2},$$

where $\hat{l} = S_{xy}/\hat{\beta}\sigma_u^2$ and $\hat{\varphi} = \hat{\sigma}_e^2 + \hat{\beta}^2\sigma_u^2$.

Proof. The result follows basically from Theorem 3.1, which implies that $\hat{l} \xrightarrow{a.s.} \frac{\nu^2}{\sigma_u^2}$ and $\hat{\varphi} \xrightarrow{a.s.} \varphi$, as $n \rightarrow \infty$.

Another consistent estimator that can be considered follows by regressing x_i on y_i in the measurement error model specified by (1.1)–(1.3). This estimator termed as the inverse estimator (see Section 2) is given by

$$\tilde{x} = \tilde{\alpha} + \tilde{\beta}\tilde{y}_0,$$

where $\tilde{\alpha} = \bar{X} - \tilde{\beta}\bar{y}$ and $\tilde{\beta} = \frac{S_{xx} - \sigma_u^2}{S_{xy}}$, $S_{xy} \neq 0$. The estimators $\tilde{\alpha}$ and $\tilde{\beta}$ can be obtained as moment estimators in the structural inverse regression of x_i on y_i , with $x_i \sim N(\mu_x, \sigma_x^2)$, $i = 1, \dots, n$. However, simple algebraic manipulations show that $\tilde{\beta} = 1/\hat{\beta}$, with $\hat{\beta}$ given as in (3.3), which implies that $\tilde{x} = \hat{x}$. Thus, under the model specified by (1.1)–(1.3), the classical and inverse estimators coincide.

4. Comparisons between the estimators

The results in Sections 2 and 3 imply, up to order n^{-1} , that all three estimators are asymptotically biased. However, as $n \rightarrow \infty$ ($k \geq 1$),

$$\text{Bias}(\hat{x}) \rightarrow 0, \quad \text{Bias}(\hat{x}_L) \rightarrow \frac{x_0 - \bar{x}}{l} \quad \text{and} \quad \text{Bias}(\tilde{x}_L) \rightarrow \frac{x_0 - \bar{x}}{1 + Z}.$$

Thus, only $\hat{x}$ is asymptotically unbiased. Note also that the three estimators are asymptotically unbiased when $\bar{x} = x_0$ and the asymptotic bias may be small when $\bar{x}$ is close to x_0 . Furthermore, as $n \rightarrow \infty$ and $k \rightarrow \infty$ the conclusions are similar. With respect to the asymptotic mean squared error, we have that

$$\lim_{n \rightarrow \infty} \text{MSE}(\hat{x}) = \frac{\sigma_e^2}{k\beta^2}, \quad \lim_{n \rightarrow \infty} \text{MSE}(\hat{x}_L) = \frac{\sigma_e^2(1+l)^2}{k\beta^2 l^2} + \frac{(x_0 - \bar{x})^2}{l^2}$$

and

$$\lim_{n \rightarrow \infty} \text{MSE}(\tilde{x}_L) = \frac{\sigma_e^2 Z^2}{k\beta^2(1+Z)^2} + \frac{(x_0 - \bar{x})^2}{(1+Z)^2}.$$

Thus, for large n , $\text{MSE}(\hat{x}) < \text{MSE}(\hat{x}_L)$. When $x_0 = \bar{x}$, it follows that $\text{MSE}(\tilde{x}_L) < \text{MSE}(\hat{x}) < \text{MSE}(\hat{x}_L)$.

4.1 Numerical evaluations of the MSEs

As seen from the expressions of their MSE for small n , it is not simple to compare analytically the three estimators. To obtain numerical comparisons between the estimators, we write the first three order approximations to their asymptotic mean squared errors as quadratic forms as considered in Lwin and Maritz (1982). After some algebraic manipulations, we write

$$\text{MSE}[\hat{x}_L] = A_1 w^2 + C_1,$$

$$\text{MSE}[\hat{x}] = A_2 w^2 + C_2$$

and

$$MSE[\tilde{x}_L] = A_3 w^2 + C_3,$$

where

$$\begin{aligned} w &= (x_0 - \bar{x}), \\ A_1 &= \frac{1}{l^2} + \frac{T(1+l)^2(l+3)}{nl^4} + \frac{(1+l^2)(l+3) - 2l(1+3l)}{nl^3(1+l)}, \\ C_1 &= \frac{\sigma_u^2}{n} \left(1 + \frac{T(1+l)^2}{l^2} \right) + \frac{T\sigma_u^2(1+l)^2}{kl^2} - \frac{2T\sigma_u^2(1+3l)}{nkl^2} \\ &\quad + \frac{3T\sigma_u^2}{nkl^3} \left(\frac{T(1+l)^3}{l} + (1+l^2) \right), \\ A_2 &= \frac{T(1+l)}{nl^2} + \frac{(l+2)}{nl^2}, \\ C_2 &= \frac{\sigma_u^2(1+T)}{n} + \frac{\sigma_u^2 T}{k} + \frac{3\sigma_u^2 T^2(1+l)}{nkl^2} + \frac{\sigma_u^2 T(2-3l)}{nkl^2}, \\ A_3 &= \frac{1}{(1+Z)^2} + \frac{1}{nT(1+Z)} + \frac{Z(Z^2 - 6Z - 1)}{n(1+Z)^4} \end{aligned}$$

and

$$\begin{aligned} C_3 &= \frac{\sigma_u^2}{n} \left(1 + \frac{TZ^2}{(1+Z)^2} \right) + \frac{TZ^2\sigma_u^2}{k(1+Z)^2} + \frac{\sigma_u^2}{nk(1+Z)} \\ &\quad \cdot \left(1 + \frac{TZ(7Z^2 + 2Z + 1)}{(1+Z)^3} \right), \end{aligned}$$

with $T = \sigma_e^2/\sigma_u^2\beta^2$. After some algebraic manipulations, it can be shown that $A_1 > 0$ if $n \geq 3$, $C_1 > 0$ if $n+k \geq 3$, $A_2 > 0$, $C_2 > 0$ and $A_3 > 0$ if $n \geq 6$ and $C_3 > 0$. The approach used to compare the three estimators is illustrated with estimators $\hat{x}_L$ and $\tilde{x}_L$. Similar comparisons hold with respect to the other pair of estimators. The following cases may arise:

- I. $C_1 > C_3$ and $A_1 > A_3$
- II. $C_1 < C_3$ and $A_1 < A_3$

III. $C_1 > C_3$ and $A_1 < A_3$

IV. $C_1 < C_3$ and $A_1 > A_3$.

In case I, $MSE[\tilde{x}_L] < MSE[\hat{x}_L]$, while in case II, $MSE[\hat{x}_L] < MSE[\tilde{x}_L]$. Further, in case III, $MSE[\tilde{x}_L] < MSE[\hat{x}_L]$ if

$$x_0 \in (\bar{x} - (\frac{C_1 - C_3}{A_3 - A_1})^{1/2}; \bar{x} + (\frac{C_1 - C_3}{A_3 - A_1})^{1/2});$$

while, in case IV, $MSE[\hat{x}_L] < MSE[\tilde{x}_L]$, if

$$x_0 \in (\bar{x} - (\frac{C_3 - C_1}{A_1 - A_3})^{1/2}; \bar{x} + (\frac{C_3 - C_1}{A_1 - A_3})^{1/2}).$$

It can be shown that $A_1 > A_2$ if $n \geq 6$ and $C_1 > C_2$ if $n + k \geq 4$. This is the situation expressed by case I above. Thus, if $n \geq 6$ it follows that $MSE[\hat{x}] < MSE[\tilde{x}_L]$.

Direct analytical comparisons between the MSE of the estimators $\hat{x}$ and $\tilde{x}_L$ and those with the MSE of $\hat{x}_L$ are complicated. Simulation studies conducted by the authors seem to indicate that $MSE[\tilde{x}_L] < MSE[\hat{x}_L]$ for small l and large β or (and) σ_u^2 . Small σ_e^2 and large β seem to indicate that $MSE[\tilde{x}_L] > MSE[\hat{x}]$ in an interval $(x_{0I}; x_{0S})$ with reversing inequality outside the interval. Otherwise, large σ_e^2 and small β seem to imply case III, that is, $MSE[\tilde{x}_L] < MSE[\hat{x}]$ in an interval $(x_{0I}; x_{0S})$ with reversing inequality outside the interval. The simulation study also shows that the distance $w = x_0 - \bar{x}$ has a great influence on the behaviour of the estimators. As expected, if w is small $\tilde{x}_L$ presents the best behaviour. Otherwise, as w increases, $\hat{x}$ behaves better in terms of mean squared error (see the simulation study in Section 4.2).

4.2 A simulation study

In this section a simulation study is presented to compare the approximated asymptotic bias and mean squared error of the estimators proposed. Since $\hat{\beta}$ may be a poor estimator when S_{XX} is closed to σ_u^2 , we propose a modified classical estimator given by:

$$\hat{x}_M = \frac{\bar{y}_0 - \hat{\alpha}_M}{\hat{\beta}_M},$$

where

$$\hat{\alpha}_M = \bar{y} - \hat{\beta}_M \bar{x} \quad \text{and} \quad \hat{\beta}_M = [\hat{H} + a(n-1)^{-1}\sigma_u^2]^{-1}S_{Xy},$$

with $a > 0$ a fixed number,

$$\hat{H} = \begin{cases} S_{XX} - \sigma_u^2, & \text{if } \hat{\lambda} \geq 1 + (n-1)^{-1} \\ S_{XX} - [\hat{\lambda} - (n-1)^{-1}] \sigma_u^2, & \text{if } \hat{\lambda} < 1 + (n-1)^{-1} \end{cases}$$

and

$$\hat{\lambda} = [S_{XX} - \frac{S_{Xy}^2}{S_{yy}}] \frac{1}{\sigma_u^2}.$$

Fuller (1987) introduces the estimator $\hat{\beta}_M$ and shows that it has better properties than the estimator $\hat{\beta}$ defined in (3.3) for small samples.

In the simulation study that follows, 1,000 samples are generated according to the model (1.1)–(1.3) considering $\alpha = 10.0$, $\beta = 0.5$, $x_0 = 15.0$, $k = 2$, $\sigma_e^2 = 1.0$, $a = 2$, $\sigma_u^2 = 0.1$ ($l = 333.3$), $\sigma_u^2 = 20.0$ ($l = 1.67$) and $\sigma_u^2 = 100.0$ ($l = 0.334$), for $n = 20, 100$. The parameters $x_1, \dots, x_n$ are generated from the uniform distribution, $U(10, 30)$. The approximated (including first order terms) asymptotic bias and mean squared error of the estimators $\hat{x}$, $\tilde{x}_L$, $\hat{x}_L$ are obtained for the parameters given above. For each selected sample, the estimators $\hat{x}$, $\tilde{x}_L$, $\hat{x}_L$ and $\hat{x}_M$ were computed and their simulated bias and mean squared error evaluated by $\sum(x_0 - \hat{x}_G)/1000$ and $\sum(x_0 - \hat{x}_G)^2/1000$, respectively, where $\hat{x}_G$ is any one of the above

estimators and $\sum$ is extended to the 1,000 generated samples. The results are presented in TABLES I, II, III and IV.

TABLE I. Bias of $\hat{x}_G$ when $n = 20$.

σ_u^2	$\hat{x}_L$		$\tilde{x}_L$		$\hat{x}$		
	empirical	approx.	empirical	approx.	empirical	empirical $\hat{x}_M$	approx.
0.1	-0.115	-0.194	0.648	0.684	-0.091	-0.094	-0.043
10.0	-2.307	-2.222	0.676	0.684	0.085	-0.336	0.162
100.0	-23.260	-23.313	0.060	0.684	-3.062	-6.818	2.027

TABLE II. Bias of $\hat{x}_G$ when $n = 100$.

σ_u^2	$\hat{x}_L$		$\tilde{x}_L$		$\hat{x}$		
	empirical	approx.	empirical	approx.	empirical	empirical $\hat{x}_M$	approx.
0.1	-0.009	-0.037	0.438	0.427	0.004	0.003	-0.005
10.0	-1.218	-1.224	0.443	0.427	0.017	-0.010	0.018
100.0	-12.647	-12.259	0.432	0.427	0.070	-0.802	0.226

TABLE III. Mean Squared Error of $\hat{x}_G$ when $n = 20$.

σ_u^2	$\hat{x}_L$		$\tilde{x}_L$		$\hat{x}$		
	empirical	approx.	empirical	approx.	empirical	empirical $\hat{x}_M$	approx.
0.1	2.548	2.523	2.206	2.256	2.521	2.524	2.498
10.0	12.126	11.260	3.311	3.283	4.426	4.438	4.125
100.0	33,652.835	954.314	13.040	12.615	490.169	1,443.825	68.817

TABLE IV. Mean Squared Error of $\hat{x}_G$ when $n = 100$.

$*\sigma_u^2$	$\hat{x}_L$		$\tilde{x}_L$		$\hat{x}$		
	empirical	approx.	empirical	approx.	empirical	empirical $\hat{x}_M$	approx.
0.1	2.097	2.081	1.856	1.831	2.084	2.085	2.069
10.0	5.128	5.132	1.985	1.980	2.232	2.257	2.241
100.0	204.351	197.848	3.272	3.329	6.935	6.489	6.849

For $\sigma_u^2 = 0.1$, note that although the bias of the inverse estimator is greater, its mean squared error is less than the mean squared error of the other estimators. In this case, the improvement of the estimator $\hat{x}_M$ was not noticed. The tables also show that, in general, the approximations are satisfactory, even for small sample sizes.

For $\sigma_u^2 = 10.0$ and $\sigma_u^2 = 100.0$, there exists some difference between the mean squared errors of $\hat{x}$ and $\hat{x}_M$ for small sample sizes, but the difference is reduced when the sample sizes increase. Note that the empirical

and approximated bias and also the empirical and approximated mean squared error of the estimator $\hat{x}_L$ can be much greater than the ones for the other estimators. Such behaviour was expected, according to the observations in Section 2. For the mean squared error, the following relation is obtained

$$MSE(\hat{x}_L) < MSE(\hat{x}) < MSE(\hat{x}_L).$$

Comparing the estimators $\hat{x}$ and $\hat{x}_M$, we can note that for $n = 20$, $MSE(\hat{x}_M) < MSE(\hat{x})$, while for $n = 100$, $MSE(\hat{x}) < MSE(\hat{x}_M)$. Fuller (1987) comments on the superiority of the modified estimator $\hat{\beta}_M$ for small sample sizes.

The percentiles of the empirical distribution of the estimators $\hat{\beta}$, $\hat{\beta}_M$, $\hat{x}$ and $\hat{x}_M$ for 1,000 samples are presented in TABLE V.

TABLE V. Percentiles of the distribution of the estimators

$\hat{\beta}$, $\hat{\beta}_M$, $\hat{x}$ and $\hat{x}_M$ with $a = 2$, $\sigma_e^2 = 1.0$,
 $\sigma_u^2 = 20.0$, $\alpha = 10.0$, $\beta = 0.5$, $x_0 = 15.0$, $k = 2$ and $n = 20$.

Percentiles	$\hat{\beta}$	$\hat{\beta}_M$	$\hat{x}$	$\hat{x}_M$
0.001	-60.2789	0.1265	-3.2206	-5.8373
0.01	-1.9253	0.2358	8.0325	7.0813
0.05	0.2950	0.2936	10.4753	9.7700
0.10	0.3394	0.3281	11.6259	10.8617
0.25	0.4189	0.3839	13.6027	12.6763
0.50	0.5226	0.4505	15.4590	14.3658
0.75	0.7242	0.5307	17.1032	15.8908
0.90	1.0594	0.5992	18.8281	17.1974
0.95	1.4812	0.6435	19.4045	17.9371
0.99	3.3991	0.7232	20.8532	18.9941
1.000	55.7968	0.8021	23.6595	20.5864

Notice that the distribution of the modified estimator $\hat{\beta}_M$ is much more concentrated than the distribution of $\hat{\beta}$. This concentration is not observed in the distribution of $\hat{x}_M$.

5. Application

In this section we consider application of the results obtained to a real data set presented in the literature. The data set considered appears in Madansky (1959), dealing with the relationship of the Brinell hardness (y) and the yield strength (x) of artillery shells. According to Madansky, it is reasonable to take the variance $\sigma_u^2 = 50$. In the sample of 50 shells reported by Madansky, the following statistics can be computed: $S_{XX} = 712.9$, $S_{Xy} = 2344.0$ and $S_{yy} = 13917.9$, leading to the following estimators: $\hat{\beta} = 3.536$, $\hat{\sigma}_e^2 = 2814.784$ and $\hat{l} = 13.258$. It can be shown that $\hat{A}_1 = 1.64 \cdot 10^{-2}$, $\hat{A}_2 = 9.04 \cdot 10^{-3}$, $\hat{A}_3 = 6.30 \cdot 10^{-2}$, $\hat{C}_1 = 271.22$, $\hat{C}_2 = 234.59$ and $\hat{C}_3 = 130.37$. Since $\hat{A}_1 > \hat{A}_2$ and $\hat{C}_1 > \hat{C}_2$; $\hat{A}_1 < \hat{A}_3$ and $\hat{C}_1 > \hat{C}_3$; $\hat{A}_2 < \hat{A}_3$ and $\hat{C}_2 > \hat{C}_3$, it follows that:

If $x_0 < 208.92$ then $\hat{x}$ is preferable to $\hat{x}_L$ which is preferable to $\tilde{x}_L$;

If $x_0 \in (208.92, 219.96)$ then $\hat{x}$ is preferable to $\tilde{x}_L$ which is preferable to $\hat{x}_L$;

If $x_0 \in (219.96, 307.88)$ then $\tilde{x}_L$ is preferable to $\hat{x}$ which is preferable to $\hat{x}_L$;

If $x_0 \in (307.88; 318.92)$ then $\hat{x}$ is preferable to $\tilde{x}_L$ which is preferable to $\hat{x}_L$;

If $x_0 > 318.92$, then $\hat{x}$ is preferable to $\hat{x}_L$ which is preferable to $\tilde{x}_L$.

Thus, unless x_0 is close to $\bar{x}$, it seems that $\hat{x}$ is the estimator to use.

6. Final discussion

In this paper, the estimation of x_0 , given the measurement error model (1.1)–(1.3), with σ_u^2 known to make the model identifiable, is considered. Several estimators for x_0 have been considered. It seems that the consistent estimator based on the estimators of α and β obtained by the method of moments under the structural model seems to be the one that presents the best behaviour unless $\bar{x}$ is close to x_0 .

Appendix

Proof of Lemma 2.1. Using standard expansions of the corresponding sample moments (see also Kendall and Stuart, 1952, 5th edition, Vol. I, pp. 322), it can be shown that

$$\begin{aligned} Var[S_{XX}] &= \frac{2\sigma_u^2}{n} [\sigma_e^2 + 2S_{xx}] + O(n^{-2}), \\ Var[S_{Xy}] &= \frac{(\sigma_e^2 + \beta^2\sigma_u^2)}{n} S_{xx} + \frac{\sigma_e^2\sigma_u^2}{n} + O(n^{-2}), \\ Var[S_{yy}] &= \frac{2\sigma_e^2}{n} [\sigma_e^2 + 2\beta^2 S_{xx}] + O(n^{-2}). \end{aligned}$$

Furthermore, simple algebraic manipulations yield

$$Cov[S_{XX}, S_{yy}] = 0, \quad Cov[S_{XX}, S_{Xy}] = \frac{2\beta\sigma_u^2}{n} S_{xx}$$

and

$$Cov[S_{yy}, S_{Xy}] = \frac{2\beta\sigma_e^2}{n} S_{xx}.$$

In addition, standard properties of the normal distribution imply that the covariance between sample means and sample variances or covariances are all zero, as for example, $Cov[\bar{y}, S_{yy}] = Cov[\bar{X}, S_{XX}] = 0$.

Notice that $\hat{\beta}_L$ and $\tilde{\beta}_L$ are continuous functions of (S_{XX}, S_{yy}, S_{Xy}) . Taking expectation of a Taylor expansion of $\hat{\beta}_L$ and $\tilde{\beta}_L$ at $a =$

$(E[S_{XX}], E[S_{yy}], E[S_{xy}])$, leads, up to first order terms, to

$$E[\hat{\beta}_L] = \frac{\beta l}{l+1-1/n} + \frac{\beta l}{\theta^2(1+l)^3} \text{Var}[S_{XX}] - \frac{1}{\theta^2(1+l)^2} \text{Cov}[S_{XX}, S_{xy}];$$

$$\begin{aligned} \text{Var}[\hat{\beta}_L] &= \frac{\beta^2 l^2}{\theta^2(1+l)^4} \text{Var}[S_{XX}] + \frac{1}{\theta^2(1+l)^2} \text{Var}(S_{xy}) \\ &\quad - \frac{2\beta l}{\theta^2(1+l)^3} \text{Cov}[S_{XX}, S_{xy}]; \end{aligned}$$

$$\begin{aligned} E[\tilde{\beta}_L] &= \frac{Z}{\beta(1+Z-1/n)} + \frac{Z}{\beta\lambda^2\theta^2(1+Z)^3} \text{Var}[S_{yy}] \\ &\quad - \frac{1}{\lambda^2\theta^2(1+Z)^2} \text{Cov}[S_{yy}, S_{xy}] \end{aligned}$$

and

$$\begin{aligned} \text{Var}[\tilde{\beta}_L] &= \frac{Z^2}{\beta^2\lambda^2\theta^2(1+Z)^4} \text{Var}[S_{yy}] + \frac{1}{\lambda^2\theta^2(1+Z)^2} \text{Var}(S_{xy}) \\ &\quad - \frac{2Z}{\beta\lambda^2\theta^2(1+Z)^3} \text{Cov}[S_{yy}, S_{xy}]. \end{aligned}$$

The result follows by replacing the expressions for the variances and covariances of the sample moments obtained above.

Proof of Theorem 2.1. With respect to $\hat{x}_L$ notice that we can write

$$\hat{x}_L = \bar{X} + \frac{\beta(x_0 - \bar{x}) + \bar{e}_0 - \bar{e}}{\hat{\beta}_L},$$

where $\bar{e} = \sum_{i=1}^n e_i/n$ and $\bar{e}_0 = \sum_{j=1}^k e_{0j}/k$. The mean and variance of $\hat{x}_L$ are given by

$$E[\hat{x}_L] = E[\bar{X}] + \beta(x_0 - \bar{x})E\left[\frac{1}{\hat{\beta}_L}\right]$$

and

$$\text{Var}[\hat{x}_L] = \text{Var}[\bar{X}] + \beta^2(x_0 - \bar{x})^2 \text{Var}\left[\frac{1}{\hat{\beta}_L}\right] + \text{Var}\left[\frac{\bar{e}_0}{\hat{\beta}_L}\right] + \text{Var}\left[\frac{\bar{e}}{\hat{\beta}_L}\right],$$

So the mean and variance depend on $1/\hat{\beta}_L$, which can be written, after a Taylor series expansion, as

$$\frac{1}{\hat{\beta}_L} = \frac{1}{E[\hat{\beta}_L]} - \frac{(\hat{\beta}_L - E[\hat{\beta}_L])}{E^2[\hat{\beta}_L]} + \frac{(\hat{\beta}_L - E[\hat{\beta}_L])^2}{E^3[\hat{\beta}_L]} + O_p(n^{-2}). \quad (A.1)$$

Using (A.1) we can write, ignoring terms of $O(n^{-2})$,

$$E[\hat{x}_L] = E[\bar{X}] + \beta(x_0 - \bar{x})\left\{\frac{1}{E[\hat{\beta}_L]} + \frac{Var[\hat{\beta}_L]}{E^3[\hat{\beta}_L]}\right\}$$

and

$$\begin{aligned} Var[\hat{x}_L] &= Var[\bar{X}] + \beta^2(x_0 - \bar{x})^2 \frac{Var[\hat{\beta}_L]}{E^4[\hat{\beta}_L]} + \frac{Var[\bar{e}_0]}{E^2[\hat{\beta}_L]} \\ &\quad + \frac{3Var[\bar{e}_0]Var[\hat{\beta}_L]}{E^4[\hat{\beta}_L]} + \frac{Var[\bar{e}]}{E^2[\hat{\beta}_L]}. \end{aligned}$$

The results follow replacing $E[\hat{\beta}_L]$ and $Var[\hat{\beta}_L]$ given in Lemma 2.1.

With respect to $\tilde{x}_L$, the proof follows by writing

$$\tilde{x}_L = \bar{X} + \tilde{\beta}_L(\beta(x_0 - \bar{x}) + \bar{e}_0 - \bar{e}),$$

and considering the required expansions, as considered in Lemma 2.1.

Proof of Theorem 3.1. First, note that $X_1, X_2, \dots$, are independent random variables with $E[X_i] = x_i$ and $Var[X_i] = \sigma_u^2$, $i = 1, \dots$. Thus, since

$$\sum_{n=1}^{\infty} \frac{Var[X_n]}{n^2} = \sigma_u^2 \sum_{n=1}^{\infty} \frac{1}{n^2} = \sigma_u^2 \frac{\pi^2}{6},$$

which is finite, it follows from Kolmogorov's strong law of large numbers that $\bar{X} - \bar{x} \xrightarrow{a.s.} 0$, as $n \rightarrow \infty$. Thus, $\bar{X}$ is a consistent estimator of $\bar{x}$.

From assumption (3.7) it then follows that $\bar{X} \xrightarrow{a.s.} \mu$ as $n \rightarrow \infty$. Similarly, it can be shown that $\bar{y} \xrightarrow{a.s.} \alpha + \beta\mu$, $S_{XX} \xrightarrow{a.s.} \sigma_u^2 + \nu^2$, $S_{Xy} \xrightarrow{a.s.} \beta\nu^2$, $S_{yy} \xrightarrow{a.s.} \sigma_e^2 + \beta^2\nu^2$, $\bar{y}_0 \xrightarrow{a.s.} \alpha + \beta x_0$ and $S_{yy}^0 \xrightarrow{a.s.} \sigma_e^2$, as $n \rightarrow \infty$. Since $\hat{\alpha}$, $\hat{\beta}$, $\hat{x}$ and $\hat{\sigma}_e^2$ are continuous functions of the above sample moments, except

when $\hat{\beta} = 0$, which has probability zero, it is easy to show that

$$\lim_{n \rightarrow \infty} (\hat{\alpha}, \hat{\beta}, \hat{x}, \hat{\sigma}_e) = (\alpha, \beta, x_0, \sigma_e^2),$$

with probability one, concluding the strong consistency of the estimators given in (3.3)–(3.6).

Proof of Lemma 3.1. Notice first that $\hat{\beta}$ is a continuous function of (S_{XX}, S_{XY}) . Taking expectation of a Taylor expansion of $\hat{\beta}$ at $\mathbf{a} = (E[S_{XX}], E[S_{XY}])$, leads, up to first order terms, to

$$E[\hat{\beta}] = \beta + \frac{\beta}{S_{xx}^2} \text{Var}[S_{XX}] + \frac{\beta \sigma_u^2}{n S_{xx}} - \frac{1}{S_{xx}^2} \text{Cov}[S_{XX}, S_{XY}]$$

and

$$\text{Var}[\hat{\beta}] = \frac{1}{S_{xx}^2} \text{Var}[S_{XY}] + \frac{\beta^2}{S_{xx}^2} \text{Var}[S_{XX}] - 2 \frac{\beta}{S_{xx}^2} \text{Cov}[S_{XX}, S_{XY}]$$

The result follows by substituting the expressions for the variances and covariances of the sample moments obtained in the proof of Lemma 2.1.

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